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Date

Abstract

Write your abstract here.

Keywords

Keyword 1, *Keyword 2*, *Keyword 3*, *Keyword 4*

Introduction / Background

Provide an introduction with background information.

Statistical Methods

Model

Suppose we define the random variable T as the number in the Vaccine group from the n = 170 Covid 19 cases

Group	Cases	Sample Size
BNT162b2	2	17411
Placebo	168	17511
Total	170	34922

The distribution of T can be well approximated by a binomial distribution which is proved in the appendix

Then,

$$T \sim Binom(n = 170, \pi)$$

Where π = P(vaccine | Covid 19 Case)

Likelihood Inference

Detail the likelihood approach.

Bayesian Inference

Under our prior beliefs,

$$P(\psi \geq 0.3) = 0.1, \quad P(\psi \geq 0) = 0.6.$$

When $\psi = 0.3$,

$$\pi = \frac{1 - 0.3}{2 - 0.3} = \frac{0.7}{1.7} = \frac{7}{17}.$$

Thus,

$$P(\psi \geq 0.3) = 0.1 \quad \Longleftrightarrow \quad P\left(\pi \leq \frac{7}{17}\right) = 0.1.$$

When $\psi = 0$,

$$\pi = \frac{1 - 0}{2 - 0} = \frac{1}{2}.$$

Thus,

$$P(\psi \geq 0) = 0.6 \quad \Longleftrightarrow \quad P\left(\pi \leq \frac{1}{2}\right) = 0.6.$$

So, our conditions for π are

$$P\left(\pi \leq \frac{7}{17}\right) = 0.1, \qquad P\left(\pi \leq \frac{1}{2}\right) = 0.6.$$

Using this to create a beta distribution gives $Beta(36.40, 38.58)$ as the prior.

```r
library(LearnBayes)
```

```
## Warning: package 'LearnBayes' was built under R version 4.4.3
```

```r
beta.select(
  quantile1 = list(p = 0.10, x = 7/17),
  quantile2 = list(p = 0.60, x = 0.50)
)
```

```
## [1] 36.40 38.58
```

Because the Beta distribution is conjugate to the Binomial, the posterior distribution of π is also a

Beta distribution.

$$T \mid \pi \sim \text{Binomial}(n, \pi)$$

and

$$\pi \sim \text{Beta}(a, b),$$

then the posterior distribution is

$$\pi \mid T = t \sim \text{Beta}(a + t,\ b + n - t).$$

For our data, $n = 170$ and $t = 8$, so the posterior distribution is

$$\pi \mid T = 8 \sim \text{Beta}(a + 8,\ b + 162).$$

Results

Present your findings.

Discussion / Conclusion

Discuss / conclude here.

Bibliography

Brown, B. (2024). *Lecture Title*. Lecture slides, Course Name, University Name.

Doe, J. (2020). Title of the Paper. *Journal Name*, 12(3), 45-67.

Last, F., & Last, F. (2025). *Book Title*. Publisher.

Smith, A., & Johnson, C. (2023). *Title of the Online Article*. Retrieved from https://www.example.com.

Appendix

Code

```
# You can reference your code in the appendix (sample here).
```

Proofs

If applicable, include detailed mathematical derivations or additional theoretical explanations.